MSc in Data Science for Economics

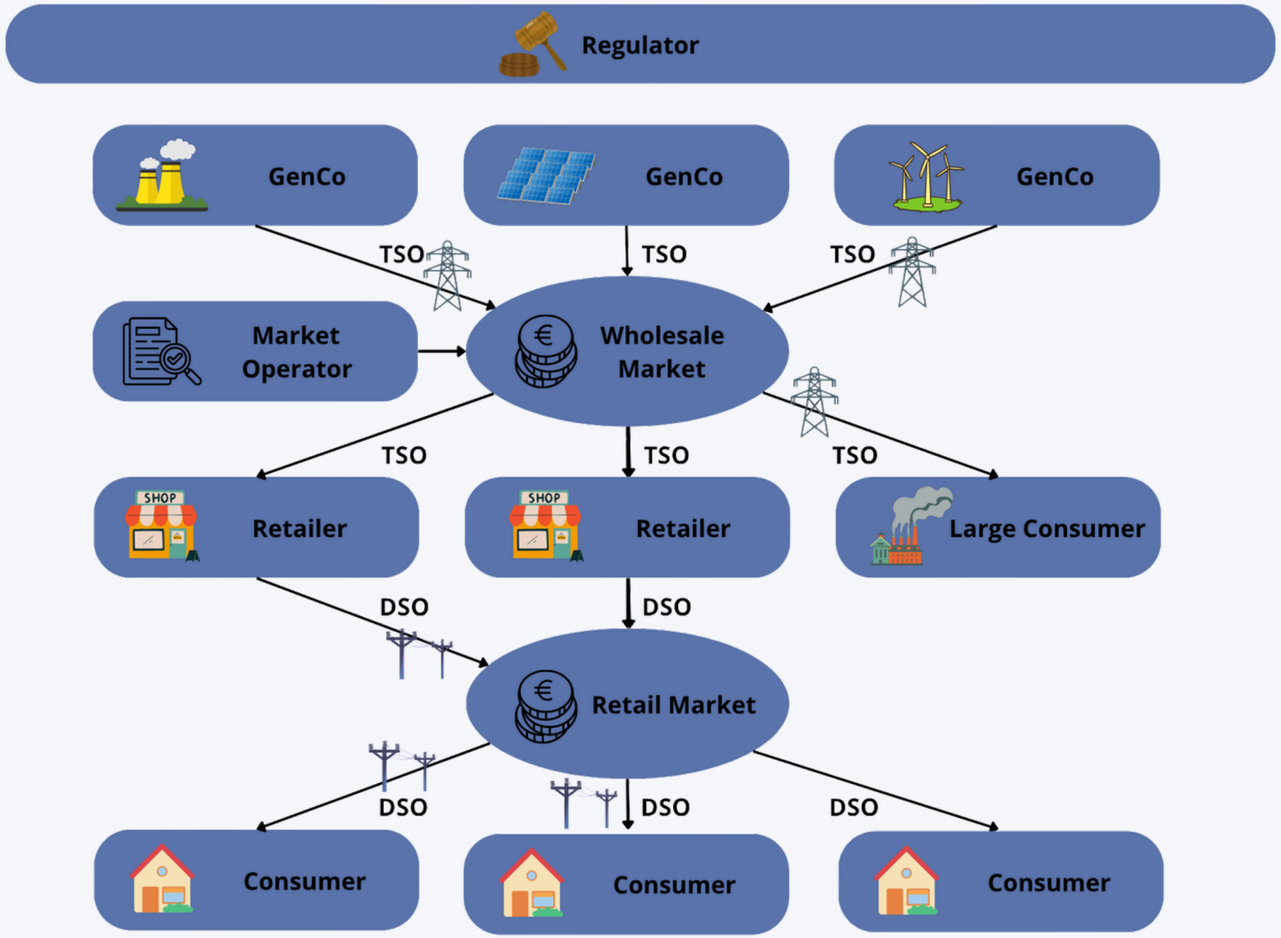
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A Structural VAR Analysis for the Italian Electricity Market

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From the Market Structure to the Merit Order



Merit Order

The equilibrium price is set by the most expensive offer needed to meet total demand

Structural VAR Model

12 Lags

RES Capacity

$$y_t = v + A_1 y_{t-1} + \ldots + A_{12} y_{t-12} + \Gamma s_t + \Lambda z_t + u_t \,, \quad u_t = B_0^{-1} w_t$$

PUN Electricity Price
TTF Natural Gas Price
Electricity Consumption
Industrial Production

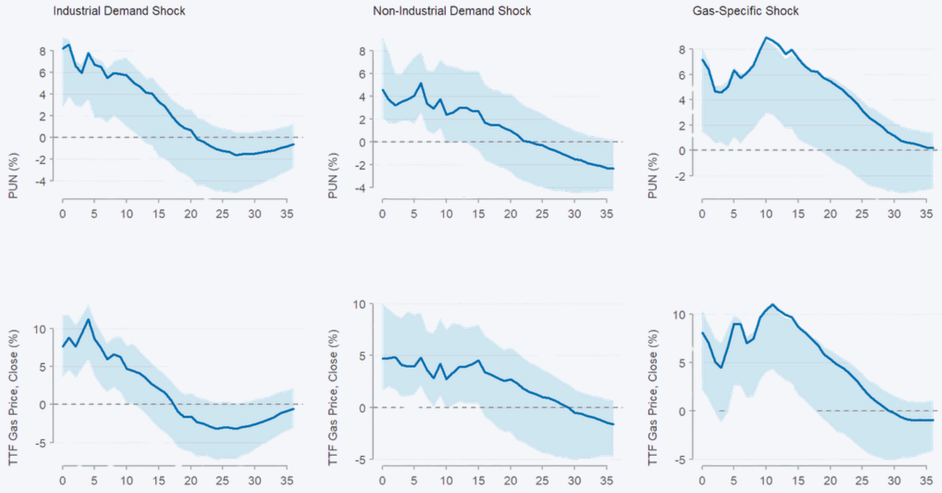
Monthly Dummies

Identification Matrix

Identification Method

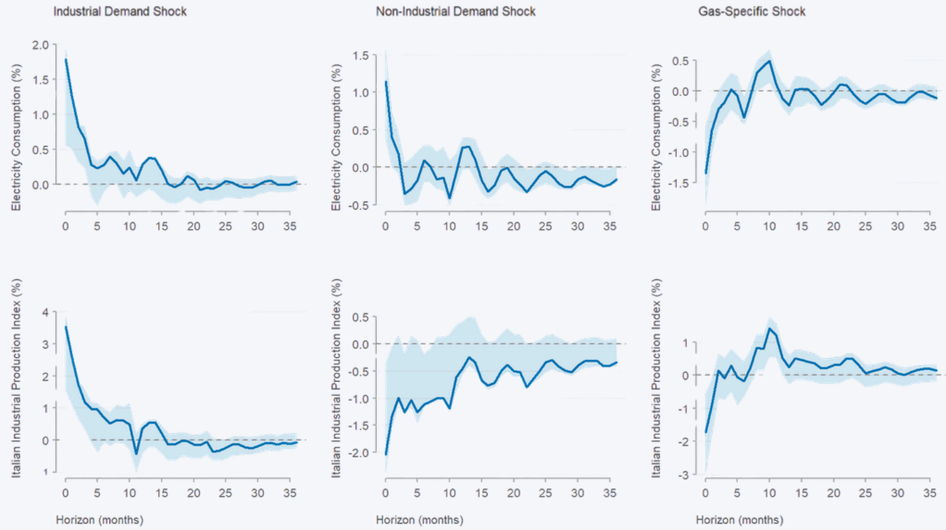
Impulse Response of	Industrial Demand	Non-industrial Demand	Gas-specific	Unlabelled
Electricity Price	+	+	+	+/-
Natural Gas Price	+	+	+	+/-
Electricity Consumption	+	+	-	+/-
Industrial Production	+	-	-	+/-

Impulse Response Functions



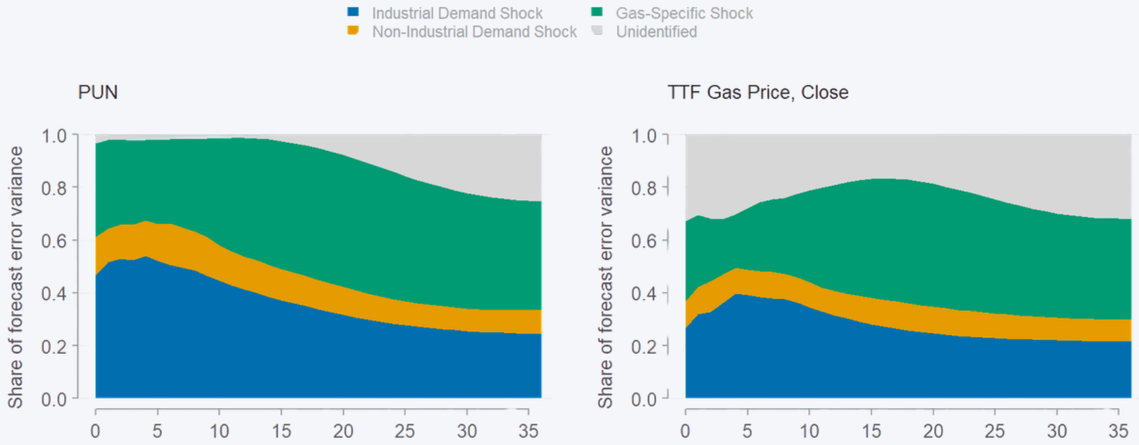
The electricity price response to all three shocks remains significant for at least ten months

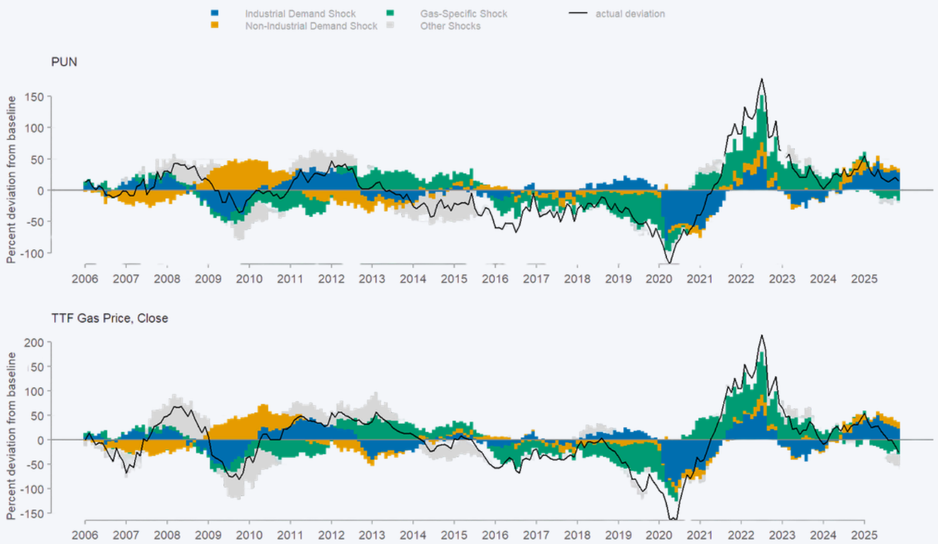
Prices respond more than quantities, both in persistence and magnitude



Historical Decomposition and Forecast Error Variance Decomposition

The gas-specific shock is the main source of PUN variance in the long run





The unlabelled shock dominates until 2016

Conditional Forecast - The Constant Gas Price



Same gas price, opposite effects on quantities

Conclusions

Main Findings

- Natural gas market shocks alone raise the PUN by 7-9%, an effect that remains significant for 18 months.
- The same price can have different underlying causes
- Prices adjust up to six times more than quantities at impact

Limitations

- Four variables may fail to capture a very complex phenomenon
- Constant parameters across different crises and scenarios
- Sign restrictions imposed only on impact leave a wide identified set

Future Work

- Focus on a shorter period of analysis
- Introduce narrative sign restrictions
- Estimate a Bayesian model with more variables

Thank you for your attention!